

Jonathan Distler

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Education

Cornell University – Ithaca, NY.....May 2027 (Expected)

• Bachelor of Science in Mechanical Engineering with a Physics Minor | GPA: 3.78

• *Relevant Coursework:*

- Math: Applied Probability and Statistics, Calculus I-III, Differential Equations, Linear Algebra
- Physics: Beginner and Intermediate Fluid Dynamics, Dynamics, Electricity and Magnetism (Honors), Heat Transfer, Mechanics, Mechanics of Engineering Materials, Statics and Mechanics of Solids, Topics in Special Relativity, Thermodynamics
- Application: Lasers and Photonics, Machine Learning for Mechanical Engineers, Mechatronics, System Dynamics

Experience

Valar Atomix Mechanical Engineering Intern

Valar Atomix – Hawthorne, CA.....May 2025 – Present

- Contributed to the successful deployment of the **world's first operational microreactor**, as part of a strategic partnership **powering NVIDIA's Blackwell AI infrastructure**
- Designed, detailed, and released **4 production components** (via Solidworks PDM) installed on the **Ward 250 reactor** during criticality
- Developed Python tools and spreadsheets to model reactor heat transfer losses
- Led **mechanical design** and structural validation of a **30' × 20' × 30', \$50K control rod drive unit (CRDU) test stand**, performing **ANSYS modal** and **static structural analyses** to assess resonance, stress, and deformation
- Completed **Radiation Worker I & II training**, authorizing entry into controlled radiological areas and higher annual dose limits

Aerospace Adversary Research Intern

Cornell University's Aerospace Adversary Lab – Ithaca, NY.....Jan 2024 – Present

- Collaborated with Dr. Gregory Falco's interdisciplinary aerospace and **cybersecurity research team**, studying autonomous **drone swarms** and **cyberwarfare** resilience
- Utilized **Linux, ROS 2, Gazebo, C++,** and **Python** to develop and test **obstacle avoidance** strategies for an autonomous 6-degree-of-freedom drone
- Developed an **Anduril-sponsored** indoor testbed, utilizing GPS with a **Python** and **ROS 2** serial communication system, to enable accurate indoor drone control with mapping precision up to **2 cm**
- Developed **Boeing-sponsored** Python simulations to localize adversarial laser sources from irradiance patterns using **eigenvalue decomposition** and **Fisher Information Matrix**-based estimation techniques
- **Co-authored** a **systems paper on autonomous drone navigation** (submitted for conference publication) ([The MAGPIE: Satellite Autonomy for Uncooperative Environments](#))

Human-Robot Collaboration and Companionship Research Intern

Cornell University's HRC2 Laboratory – Ithaca, NY.....Oct 2025 – Present

- Collaborated with Dr. Guy Hoffman's advanced robotics research team, studying human-robot interactions, designing **novel tests** for **analyzing communication modes, failure recovery,** and **cooperative task strategies**
- Designed a custom **task-domain definition** and implemented **A* search with a Fast-Forward heuristic**, generating optimized common-sense task sequence in **0.11 seconds**
- Developed **ROS 2** and **Python packages** for the **Hello-Robot Stretch RE1**, enabling autonomous navigation using **SLAM, LiDAR-based mapping,** and **Intel RealSense sensing**
- Used Boris software to manually code and annotate **80+ robotic trial videos** using predefined behavioral classifications, including **success, failure,** and **lapse** events

Soft Robotic Fish (SoFi) Research Intern

ETH Zurich's Soft Robotics Laboratory – Zurich, Switzerland.....May 2025 – Aug 2025

- Collaborated with Dr. Robert Katzschman's lab to develop a novel **actuation mechanism** and **autonomous control** for a soft robotic fish tail utilizing a Scotch yoke assembly and a novel radio-frequency methodology under \$80
- Developed a one-to-one **MuJoCo simulation** with **Nelder-Mead optimization**, validating results within 10% of experimental measurements
- Designed and implemented a **motor and IMU control class** to track fish orientation using Euler angles, integrating a Dynamixel motor and Adafruit IMU with sensor fusion and data wrapping techniques
- Implemented **computer vision** with **remote filming** to measure tail curvature, comparing results with thrust and motor data to optimize tail design and motor actuation rate

Cornell University Hyperloop

Aeroshell Subteam Lead – Ithaca, NY.....May 2026 – Present

- Applied the **Navier–Stokes momentum equations** and **CFD simulations** to **optimize the aeroshell geometry**, reducing aerodynamic drag and improving flow streamlining
- Performed **finite element analysis (FEA)** to evaluate **structural response** under aerodynamic and operational loading, verifying design safety factors and identifying high-stress regions
- Co-developed a **strip-planking manufacturing process** to **fabricate a full-scale negative mold** of the optimized aeroshell, approximating complex curvature with discrete wooden planks.
- **Collaborated** with **ETH Zurich's Swissloop** team to verify structural assumptions and support the fabrication of a strip-planked negative mold for the aeroshell

Structures Subteam Member – Ithaca, NY.....Sept 2024 – May 2026

- Performed **shear force analysis of the Hyperloop chassis** using known geometry and point-load conditions to evaluate structural integrity
- **Derived DC motor torque requirements for static launch**, accounting for static friction and transition to linear induction motor propulsion
- Developed a Python script to fuse data from four time-of-flight sensors, **estimating the vehicle's center of mass, yaw, pitch, and roll** in real time
- Co-developed a **PID controller for the electromagnetic levitation system** to regulate induced magnetic forces and maintain vehicle attitude during competition

Biomechanics Research Intern

Virginia Tech Department of Mechanical Engineering – Blacksburg, VA.....June 2024 – Aug 2024

- Developed an **H-Bot control system** using **MATLAB** and **serial communication** to synchronize **ultrasound imaging of octopode movements** for enhanced scan accuracy
- Engineered a **soft-robotic muscle prototype** with controlled expansion and contraction, performing material property analysis as part of a research team
- Developed a novel, and **entirely soft McKibben-Actuation tendon** with an **expansion rate** greater than **150%**, costing less than **\$20**.

Human-Drone Interaction Research Intern

Virginia Tech Department of Mechanical Engineering – Blacksburg, VA.....Dec 2023 – June 2024

- Co-developed a **conceptual framework for a mass drone delivery system** to address **food insecurity** within **rural communities**
- **Co-authored a paper accepted and presented** at the **IEEE Region 8 MIPRO Conference** (Exploring the Utilization of Drone Technology to Promote Food Security)

Stormwater Maintenance Personal Assistant

Stormwater Solutions – Blacksburg, VA.....Aug 2019 – Aug 2024

- Applied civil and environmental engineering techniques to **maintain bioretention ponds** and **mitigate local flooding** as part of the **Blacksburg Town Council's Stormwater Initiative**
- Enhanced existing bioretention ponds to ensure full compliance with all town-sanctioned inspections, achieving a **100% compliance rate**
- Created a locally hosted **HTML website to improve customer experience**, showcasing StormWater Solution's licenses and certifications, and providing examples of previous work, with included price-estimates

Computer Science Genome Research Intern

Virginia Tech Department of Computer Science – Blacksburg, VA.....May 2022 – Sept 2022

- Showed data trends amongst the **500 fastest benchmarked computers in the world**, creating an interactive Python notebook that used **JSON** and **API** to retrieve expansive datasets
- Using data analysis skills, I was able to derive data and trends from pulled data sets and successfully produce findings for the **CS Genome Project's website**

Awards and Recognition

- Cornell Jiang Fellows Cohort (2026)
- Cornell Tatum Family Student Career Award (2025)
- Cornell Engineering International Summer Research Grant (2025)
- Cornell Switch the Pitch Hackathon: AI Technical Solution Award Winner (2023)

Conference Publications

- *The MAGPIE: Satellite Autonomy for Uncooperative Environments*, Hawaii International Conference on Space System, 2025
- *Exploring the Utilization of Drone Technology to Promote Food Security*, IEEE Region 8 MIPRO Conference, 2024

Campus Involvement

- *Nuclear is Clean Energy (NiCE) Member*; Cornell University.....Aug 2025 - Present
- *Cornell Brazilian Jiu Jitsu Club Member*; Cornell University.....Jan 2025 - Present

Volunteering

- Future Civilization Conference Volunteer – 2026
- Cornell University Move-In Volunteer – 2024, 2025, 2026
- Northeast Robotics Colloquium Volunteer (NERC) – 2025
- Virginia Science Olympiad (VASO) Division C Material Science Test-Writer – 2024